

EXAM 7 – SPRING 2013

9. (3.5 points)

An actuary is considering a Bayesian approach to developing a predictive loss distribution based on the deterministic chain-ladder method using the all-years weighted average link ratios.

a. (0.5 point)

To produce results that closely resemble the deterministic chain-ladder outcome, explain whether the actuary should select high or low variances for the prior distributions of the link ratios.

b. (0.5 point)

The actuary decides to override the link ratios suggested by the data for the 36-48 month maturity interval with a judgmental selection. However, the actuary is less confident of this selection than of the all-years weighted averages used for the other maturity intervals. Describe how to change the prior distributions of the link ratios to adjust for this.

c. (1 point)

Discuss the effect of the change implemented in part b. above on the simulated results, addressing both the mean and the prediction error.

d. (1 point)

Describe one advantage that a Bayesian approach has over a bootstrapping algorithm and one advantage that a Bayesian approach has over the Mack method.

e. (0.5 point)

Discuss a modification to the Bayesian framework for the chain-ladder method so that it applies to the Bornhuetter-Ferguson method.

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Question 9 Sample Answer

Solution 1

- a) High Variances – If we are not confident in our prior estimates, a wide variance will cause the model to output parameters based on the actual data rather than our prior opinion.
- b) Let the distribution for the LDF_{36-48} have the actuary's opinion as the expected value, but leave a wide variance around the estimate. This will let the model consider the actuary's selection to some degree, but will still use the historical data to determine the parameter.
- c) It will pull the LDF_{36-48} closer to the actuary's estimate. Because the LDF's distribution specified a large variance, prediction error will be similar to the chain ladder, though probably larger as the variance selection is large.
- d) Over bootstrapping: Can insert your opinion into the parameter selection without much difficulty. Over Mack: We will get a full distribution of the loss estimate, not just the first two moments.
- e) We could insert row parameters, one for each accident year, and specify relatively tight variance around them. This will make the model use the prior estimate more, as in BF where the reserve estimate is based on our prior expected LR.

Solution 2

- a) Use large variance for the prior distribution to put more weight on the chain ladder outcome. The larger variance reflects we are not as confident in our prior distribution.
- b) Put a distribution with a mean equal to his selection, but with large variance. For example, for the 36-48 interval LDF,
 $\gamma' = 1.5$, $\text{variance}(\gamma') = w$; such that w with large variance
- c) The simulated results will incorporate the expert opinion for the 36-48 LDF, and the prediction error will be higher since we are less confident in our selection for this LDF than the weighted average.
- d) Compared to bootstrapping, the Bayesian approach can incorporate expert knowledge into the selection of the ratios. Compared to Mack, the full predictive distribution can be easily calculated and we can calculate the prediction error as the square root of the MSE of the distribution.

- e) If we use really strong priors (i.e. low variance) for the row parameters, this allows us to set row parameters equal to the BF estimate of ultimate for each year. This will replicate the BF method.

Examiner Comment

- a) Responses with low variance were not given credit.
- b) In order to receive full credit candidate responses had to include the use of high variance for the 36-48 LDF interval due to the actuary's high uncertainty with the selected LDF.
- c) In order to receive full credit candidate responses had to discuss the effect of the change in b. on mean and predication error of the simulated results. A candidate's response may receive part c. credit even if they did not receive full credit for part b.
- d) Candidates were given full credit for response which describing the same advantage of Bayesian over both. For example full credit could be given for response which identified and described the accommodation of actuary's expert opinion in the predictive distribution for reserves afforded by Bayesian approach is an advantage over both Mack and Bootstrapping.
- e) Overall this question part appeared to be the most difficult for candidates. Candidate's responses frequently discussed the BF modification to the chain ladder method but did not address predictive or stochastic features. Credit was not given to candidates who only discussed the deterministic, non-stochastic BF model modification to the chain ladder.